

Functional Requirements

1 Client Functional Requirements

1.1 View Shops - Alexandra

The system shall display a list of all available airport shops.

Each shop shall display:

shop name

shop type

The client shall be able to select exactly one shop at a time.

After selecting a shop, the system shall display all items available in that shop.

Verification:

Select a shop and confirm that only that shop's items are displayed.

1.2 Search Items - David

The system shall provide a search bar after a shop is selected.

The client shall be able to search for items by name.

The system shall display all items whose name contains the entered text (case-insensitive).

If no items match, the system shall display a “No items found” message.

Verification:

Search using partial and full item names.

1.3 Sort Items - Maya

The client shall be able to sort displayed items by:

name (ascending)

price (ascending)

Only one sorting criterion may be active at a time.

Verification:

Apply sorting options and confirm order.

1.4 Manage Cart - Andrada

The client shall be able to:

add items to the cart

remove items from the cart

change item quantity

The system shall prevent adding quantities greater than available stock.

The cart shall store:

item name

unit price

quantity

total price per item

Verification:

Attempt to add more items than available stock.

1.5 View Cart - Maya

The client shall be able to view cart contents at any time.

The cart view shall display:

list of items

quantity per item

individual item total

total cart price

Verification:

Add items and confirm totals are correct.

1.6 Reserve Cart - David

The client shall be able to reserve all items currently in the cart.

Reservation shall only succeed if sufficient stock exists.

When a reservation is confirmed:

stock quantities shall decrease

the reservation shall be stored in the system

The client shall be able to cancel a reservation.

Upon cancellation:

stock quantities shall be restored

the cart shall be cleared.

Verification:

Reserve items and confirm that stock quantities decrease accordingly.

Cancel the reservation and confirm that stock quantities are restored and the cart is cleared.

1.7 Flight-Based Discount - Alexandra & WizzErr(Ticket Management)

The system shall retrieve flight information from the ticket management system.

If the time difference between two flights is greater than 4 hours, the client shall receive a 10% discount.

The discount shall apply only to grocery shops.

The system shall display when a discount is applied.

Verification:

Simulate flight time differences below and above 4 hours and verify that the discount is applied only when the condition is satisfied.

2 Administrator Functional Requirements

2.1 Manage Shops - Andrada

The administrator shall be able to:

add a shop

update a shop

delete a shop

Each shop shall include:

name (unique)

type

Deleting a shop shall remove its inventory.

2.2 Manage Global Items - Felician

The administrator shall be able to manage the global product catalog.

Each item shall include:

name

description

category

The administrator shall be able to:

add items

update items

delete items

2.3 Manage Shop Inventory - Denisa

The administrator shall select a shop before managing inventory.

The administrator shall be able to:

add an item to a shop

remove an item from a shop

update item price

update stock quantity

The system shall ensure:

stock ≥ 0

price > 0

2.4 View Shop Ratings - Alex & Cloud Spritzers (Customer Support)

Each shop shall have a rating based on customer reviews.

The administrator shall be able to view:

average rating

number of reviews

2.5 Sort Shops - Alex

The administrator shall be able to sort shops by:

name (ascending)

rating (descending)

Only one sorting criterion may be active at a time.

Verification:

Apply each sorting option and confirm the correct ordering of shops.

3 Cross-System Requirements

The system shall integrate with the ticket management system to retrieve passenger flight data.

The system shall integrate with the customer support system to store and display shop reviews.

4 Non-Functional Requirements

The system shall prevent invalid input such as negative prices or stock values.

The system shall respond to user actions within 2 seconds under normal load.

The system shall maintain data consistency after reservations and cancellations.

The system shall support concurrent reservation without inconsistent stock updates.

The system shall maintain data persistence for shops, items, and reservations.